

Computer Networks



Overview of the Internet

Protocol layering (Book 1 Ch 1)

Application Layer

- Application-Layer Paradigms
- Client-server applications
 - World Wide Web and HTTP
 - FTP
 - Electronic Mail
 - DNS
- Peer-to-peer paradigm
 - P2P Networks
- Case study: BitTorrent (Book 1 Ch 2)

Protocol Layering - Introduction

- A **protocol** defines rules for communication
- Rules must be followed by:
 - Sender
 - Receiver
 - Intermediate devices
- Goal:
 - Effective communication
- Simple communication:
 - One simple protocol may be enough
- Complex communication:
 - Tasks divided into multiple layers
 - Each layer uses its own protocol
- This approach is called:
 - **Protocol Layering**

Scenario 1 - Simple Communication

- Communication between:
 - Maria
 - Ann
- Both are neighbors
- Communication characteristics:
 - Face-to-face
 - Same language
 - Single communication layer
- Communication is direct and simple

A single-layer protocol

Figure 1.9 *A single-layer protocol*



Rules Followed in Scenario 1

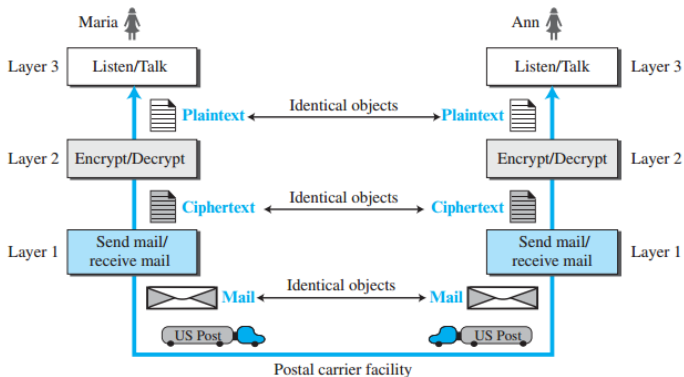
- Maria and Ann follow communication rules:
 - Greet each other
 - Use suitable vocabulary
 - Avoid speaking simultaneously
 - Maintain dialog communication
 - Exchange polite closing words
- Protocol depends on communication context
- Example:
 - Professor-student communication differs
- Classroom communication:
 - Mostly monologue
 - Formal communication
 - Student raises hand before speaking

Scenario 2 - Need for Multiple Layers

- Ann moves to another city
- Maria and Ann continue communication
- Communication medium:
 - Regular postal mail
- Concern:
 - Letters may be intercepted
- Solution:
 - Encryption and decryption
- Sender:
 - Encrypts message
- Receiver:
 - Decrypts message
- Communication now uses:
 - Three layers

A three-layer protocol

Figure 1.10 A three-layer protocol



Three-Layer Communication Process

- Maria uses three machines (robots)
- Layer 3:
 - Creates plaintext message
 - English letter format
- Layer 2:
 - Encrypts plaintext
 - Produces ciphertext
- Layer 1:
 - Places ciphertext in envelope
 - Adds sender and receiver addresses
 - Mails the letter



Receiving Process at Ann's Side

- Layer 1:
 - Receives letter from mailbox
 - Identifies sender
 - Extracts ciphertext
- Layer 2:
 - Decrypts ciphertext
 - Produces plaintext
- Layer 3:
 - Reads plaintext
 - Interprets message
- Communication behaves as if:
 - Maria is directly speaking to Ann

Advantages of Protocol Layering

- Complex tasks divided into:
 - Smaller tasks
 - Simpler tasks
- Supports:
 - Modularity
- Only affected layer needs modification
- Example:
 - Change encryption layer only
- Other layers remain unchanged
- Layers operate independently

Layer as a Black Box

- A layer can be treated as:
 - A black box
- Focus only on:
 - Inputs
 - Outputs
- Internal implementation is irrelevant
- Different manufacturers can create compatible machines
- Requirement:
 - Same output for same input
- Example:
 - Different encryption machines producing same ciphertext

Separation of Services and Implementation

- Protocol layering separates:
 - Services
 - Implementation
- Each layer:
 - Receives services from lower layer
 - Provides services to upper layer
- Internal implementation is unimportant
- Example:
 - Maria may manually perform first-layer tasks
- Communication works if services are maintained

Importance in Internet Communication

- Internet communication involves:
 - End systems
 - Intermediate systems
- Intermediate systems may require:
 - Only some layers
- Without layering:
 - Intermediate systems become very complex
- Layering reduces:
 - Cost
 - Complexity

Disadvantages of Protocol Layering

- Single-layer systems may appear simpler
- No need for:
 - Inter-layer service coordination
- Example:
 - One machine performing all tasks
- Problem:
 - Entire machine must be replaced if one function fails
- Layered systems:
 - Require replacement of only affected layer

Summary

- Protocols define communication rules
- Complex communication requires protocol layering
- Layering divides tasks into manageable modules
- Major advantages:
 - Modularity
 - Flexibility
 - Scalability
 - Easier maintenance
- Layers provide services to adjacent layers
- Internet architecture heavily depends on layering



Principles of Protocol Layering

- Protocol layering follows important design principles
- Principle 1:
 - Bidirectional communication requires dual functionality
 - Each layer must perform opposite tasks
 - One task in each direction
- Examples:
 - Layer 3:
 - Listen
 - Talk
 - Layer 2:
 - Encrypt
 - Decrypt
 - Layer 1:
 - Send mail
 - Receive mail

Principle of Identical Objects Between Layers

- Principle 2:
 - Objects under corresponding layers must be identical
 - Same type of data must exist at both communication sites
- Examples:
 - Layer 3 object:
 - Plaintext letter
 - Layer 2 object:
 - Ciphertext letter
 - Layer 1 object:
 - Piece of mail
- Ensures:
 - Compatibility
 - Proper communication between corresponding layers



Logical Connections in Protocol Layering

- Protocol layering creates:
 - Layer-to-layer communication
- Each corresponding layer communicates logically
- Logical connection means:
 - Imaginary communication link
 - Exists between same layers at both sites
- Maria and Ann assume:
 - Each layer can directly exchange its objects
- Examples of exchanged objects:
 - Plaintext
 - Ciphertext
 - Mail
- Logical connections simplify understanding of:
 - Data communication
 - Networking
 - Protocol layering tasks

Logical connection between peer layers

Figure 1.11 *Logical connection between peer layers*

